

Name: _____

Roll Number: _____

Quiz-3

Max. Time: 20 min

Max. Points: 20

Note: Solve all parts. Limit your written responses to the provided space.

Q.1. [8] Choose by putting a check mark on the most appropriate option. Note: No cutting/overwriting is allowed.

i. Every linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a matrix transformation.

☒ (A) True ☐ (B) False

ii. A transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ with A as its standard matrix, is onto if and only if the columns of A span $\mathbb{R}^m$.

☒ (A) True ☐ (B) False

iii. The columns of the standard matrix for a linear transformation A from $\mathbb{R}^m$ to $\mathbb{R}^n$ are the images of the columns of $n \times n$ identity matrix I .

☐ (A) True ☒ (B) False

iv. When two linear transformations are performed one after another, the combined effect may not always be linear.

☐ (A) True ☒ (B) False

v. A linear map represented by A , which is a 3×5 matrix, is an onto function.

☒ (A) True ☐ (B) False

vi. The codomain of a transformation $T: \mathbf{x} \mapsto A\mathbf{x}$ is always the set of all linear combinations of the columns of A .

☐ (A) True ☒ (B) False

vii. The transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is onto $\mathbb{R}^m$ if every vector $\mathbf{x}$ in $\mathbb{R}^n$ maps onto some vector in $\mathbb{R}^m$.

☐ (A) True ☒ (B) False

viii. If $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation and $\mathbf{c}$ is in $\mathbb{R}^m$, then whether $\mathbf{c}$ is in the range of T is existence question.

☒ (A) True ☐ (B) False

Q.2. [5+5]

a) Show that the following transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, which transforms $\mathbf{e}_1$ and $\mathbf{e}_2$ to $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$ respectively, is linear.

Solution:

$$T(\mathbf{e}_1) = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \text{ and } T(\mathbf{e}_2) = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \text{ therefore } T(\mathbf{x}): \mathbf{x} \mapsto A\mathbf{x}, \text{ where } A = \begin{bmatrix} 1 & 3 \\ -2 & 1 \end{bmatrix}$$

To show linearity of T , we need to show the following:

$$\begin{aligned} 1. \quad & \text{If } \mathbf{x} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, A\mathbf{x} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ & \Rightarrow \begin{bmatrix} 1 & 3 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{aligned}$$

$$2. \quad \text{For any vectors } \mathbf{u} \text{ and } \mathbf{v} \in \mathbb{R}^2, A(\mathbf{u} + \mathbf{v}) = A(\mathbf{u}) + A(\mathbf{v}).$$

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$$\text{Let } \mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \text{ and } \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \Rightarrow \mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix}$$

$$\Rightarrow A(\mathbf{u} + \mathbf{v}) = \begin{bmatrix} 1 & 3 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix} = \begin{bmatrix} u_1 + v_1 + 3u_2 + 3v_2 \\ -2u_1 - 2v_1 + u_2 + v_2 \end{bmatrix} = \begin{bmatrix} u_1 + 3u_2 + v_1 + 3v_2 \\ -2u_1 + u_2 - 2v_1 + v_2 \end{bmatrix}$$

$$= \begin{bmatrix} u_1 + 3u_2 \\ -2u_1 + u_2 \end{bmatrix} + \begin{bmatrix} v_1 + 3v_2 \\ -2v_1 + v_2 \end{bmatrix} = A(\mathbf{u}) + A(\mathbf{v})$$

b) Consider $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$. Give the standard matrix for the linear transformation A that first rotates points through the origin and then reflects them through the line $x_2 = -x_1$.

Solution:

Let $T_1 = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$ and $T_2 = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$ be the rotation and reflection, respectively. Then

$$A = [T_2(T_1(e_1)) \quad T_2(T_1(e_2))]$$

$$A = [T_2\left(\begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix}\right) \quad T_2\left(\begin{bmatrix} -\sin(\theta) \\ \cos(\theta) \end{bmatrix}\right)]$$

$$A = [T_2\left(\begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix}\right) \quad T_2\left(\begin{bmatrix} -\sin(\theta) \\ \cos(\theta) \end{bmatrix}\right)]$$

$$A = \begin{bmatrix} -\sin(\theta) & -\cos(\theta) \\ -\cos(\theta) & \sin(\theta) \end{bmatrix}$$

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